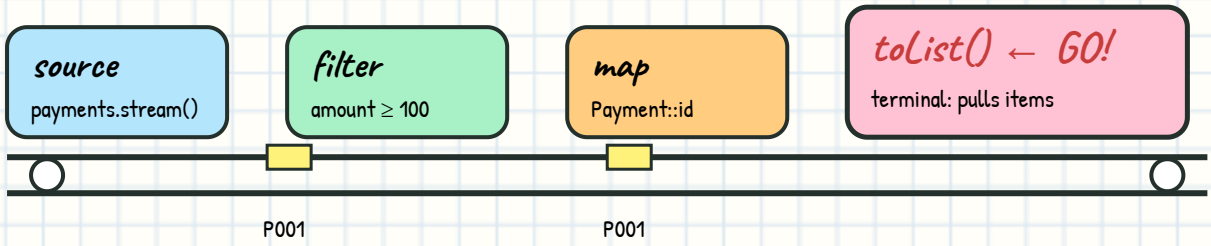
Streams · my notes L06

a conveyor belt, not a box: it describes work and stores nothing

The belt: source → steps → terminal (the GO button)

no terminal → nothing runs at all (withoutTerminal prints no trace)



One item at a time ↓

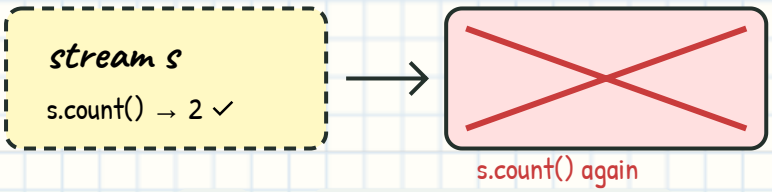
filter P001 · map P001
filter P002 × 45.50 < 100, dropped
filter P003 · map P003 · limit(2) full → STOP
= vertical order (loop fusion): one pass, no temporary lists

sorted() = a dam 🌊

water piles up until ALL items arrive:
filter P001…P005 first,
then map P001, P003, P005

stateful barrier

Use it once, like a train ticket 🎫



IllegalStateException:
"stream has already been operated upon or closed"

fix → keep the `List` or a `Supplier<Stream<Payment>>` · the list still has 12 payments

Order of operations matters

× limit(3) → sorted

takes P001, P002, P003 by **position**,
then sorts only those 3

✓ sorted → limit(3)

top 3 debits:
T042 1432.93 · T057 962.80 · T050 903.91

Where streams come from

`list.stream()` 4 GOLD customers

`IntStream.rangeClosed(10, 12)` Q4 months

`iterate(seed, f)` [8] ♾ needs limit

`iterate(seed, test, f)` [9] = for loop

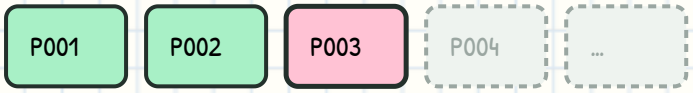
`Stream.generate(s)` ♾ needs limit

`Stream.ofNullable(x)` [9] 0 or 1

`"PAYPAL".chars()` 'P' × 2

`Files.lines(csv)` in try → 8 SUCCESS

Short-circuit = stop as soon as you know



anyMatch(FAILED) → true after 3

```
iterate(1, n → n*2)
.filter(n > 1000).limit(3)
```

→ [1024, 2048, 4096] ✓ ends

Method map doodle

- **stateless:** filter, map, flatMap, peek, mapMulti [16]
- **stateful:** distinct, sorted (barriers) · limit, skip, take/dropWhile (keep a count or flag)
- **short-circuit:** limit, takeWhile [9], anyMatch, findFirst
- **terminal:** forEach, collect, reduce, count, toList() [16]
- gather: Beyond Java 21 (Java 24)

Properties

- List → ORDERED ✓ · HashSet → ×
- `parallel().sequential()` → sequential: last call wins
- order of ops matters: `sorted` then `limit` = top 3 debits

★ Remember

lazy until the terminal · one item at a time · sorted waits for everyone · use once · infinite needs a stop · source untouched · top debit `T042 1432.93` · next → L07 filtering, mapping & slicing